UTTARANCHAL UNIVERSITY

Faculty of Uttaranchal Institute of Technology (UIT)

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Syllabus

Course : B.TECH. **Year/Sem :** 7 **Branch :** CSE

Subject : DIGITAL IMAGE PROCESSING AND PATTERN RECOGNIZATION

Unit	Sub Unit	Topic
Unit 1: Introduction to Image & Pattern Recognition		Pixels, Intensity, Coordinate Conventions, Basics of pattern recognition. , Data sets for Pattern Recognition, Different Paradigms of Pattern Recognition. , Learning and adaptation, Pattern recognition approaches. , Metric and non-metric proximity measures, Mathematical foundations - Linear algebra. , Probability Theory, Expectation, mean and covariance. , Normal distribution, multivariate normal densities. , Chi squared test and Unit 1 Review. , Design principles of pattern recognition system, Representations of Patterns and Classes.
Unit 2: Statistical Pattern Recognition		Introduction to Statistical Pattern Recognition and Bayesian Decision Theory. , Bayesian Decision Theory (Continuous features, two-category classification). , Classifiers and Decision Surfaces. , Maximum likelihood method (Theory and Formulation). , Maximum likelihood method (Applications and Problem Solving). , Normal density estimation. , Discriminant functions for normal density. , Linear Discriminant Function and Unit 2 Review.
UNIT 3: Image Preliminaries and Processing		Image Acquisition: Energy, optical system, image sensor, digital image formation, Gray scale and color images. , Feature extraction, Different approaches to Feature Selection. , Image Point Processing: Gray-level mapping, non-linear gray level mapping. , Image histogram, histogram stretching, histogram equalization, histogram matching, thresholding. , Neighborhood Processing: Median filter, mean filter, correlation, template matching. , Edge detection and image sharpening. , Color image processing. , Morphology: Dilation & Erosion, closing & opening, boundary detection.
UNIT 4: Feature Extraction and Dimension Reduction		Color, Texture, Shape, Local Features. , Spatial and frequency domain, HOG, Corner Detection. , SIFT and SURF. , Hough Transform, Parameter estimation methods introduction. , Dimension reduction methods: Principal Component Analysis (PCA). , Fisher Linear discriminant analysis. , Expectation-maximization (EM). , Hidden Markov Models (HMM), Gaussian mixture models.
UNIT 5: Nonparametric Techniques of Pattern Recognition		Introduction to Nonparametric Techniques, Density Estimation fundamentals. , Density Estimation (Advanced evaluation and error bounds). , Parzen Windows (Theory and formulation). , Parzen Windows (Implementation and practical considerations). , K-Nearest Neighbor Estimation (Theory). , K-Nearest Neighbor Estimation (Metrics and computational complexity). , Nearest Neighbor Method. , Comparative analysis of classifiers and Course Wrap-up.
Zero PPT		ZERO PPT

Recommended References

1) Legal Research Methodology, Dr. S.R. Myneni,
(T1)

2)Legal Education in India- Challenges and Perspectives, Prof. G. Manohar Rao, Prof. K. Shrinivas Rao,(R1)